THE NUMBER OF TOP-DIMENSIONAL MINIMAL PRIMES OF A NOETHERIAN LOCAL RING IS BOUNDED BY THE LENGTH OF THE QUOTIENT BY A SYSTEM OF PARAMETERS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The statement writes R/(g_1, ..., g_n) although the system of parameters has d entries, so the symbol n does not denote the dimension; that is the author's text and is reproduced unchanged. The author also writes 'of degree exactly(!) d' inside the counted proof span, reproduced unchanged. The six references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Hilbert functions of an ideal in a local ring and the degree of the polynomial giving the length of a quotient. used at proof lines 38330-38343. Is the asymptotic tool that turns the support computation into a numerical bound.
- Length additivity in a short exact sequence of finite modules over a local ring. invoked at proof lines 38333-38338. Is the comparison that relates the three Hilbert polynomials.
- The multiplicity lemma bound relating the leading coefficient to the length. invoked at proof lines 38344. Converts the leading-coefficient lower bound t/d! into the asserted length inequality.
- The one-dimensional case of the statement. invoked at proof line 38306. Is the base case of the reduction.

Source reference. Original lemma 00KC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 746–746. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00L8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 778–778. Cross-chapter navigation pointer into the

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 38016: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38016–38016. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 38261: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38261–38261. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension d, let t be the number of minimal prime ideals of R of dimension d, and let $(g_1, \ldots, g_d)$ be a system of parameters. Then $t \leq length_R(R/(g_1, \ldots, g_n))$.*

Proof. If $d = 0$ the lemma is trivial. If $d = 1$ the lemma is Lemma 38016. Thus we may assume $d > 1$. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_s$ be the minimal prime ideals of R where the first t have dimension d, and denote $I = (g_1, \ldots, g_n)$. Arguing in exactly the same way as in the proof of Lemma 38016 we can assume R is reduced.

Assume R is reduced with minimal primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_t$. This means there is an exact sequence

$$0 \to R \to \prod\nolimits_{i=1}^{t} R/\mathfrak{p}_i \to Q \to 0$$

Here Q is the cokernel of the first map. Write $M = \prod_{i=1}^{t} R/\mathfrak{p}_i$. Localizing at $\mathfrak{p}_j$ we see that

$$R_{\mathfrak{p}_j} \to M_{\mathfrak{p}_j} = \Big(\prod\nolimits_{i=1}^{t} R/\mathfrak{p}_i\Big)_{\mathfrak{p}_j} = (R/\mathfrak{p}_j)_{\mathfrak{p}_j}$$

is surjective. Thus $Q_{\mathfrak{p}_j} = 0$ for all j. Therefore no height 0 prime of R is in the support of Q. It follows that the degree of the numerical polynomial $n \mapsto \text{length}_R(Q/I^nQ)$ equals $\dim(\text{Supp}(Q)) < d$, see Algebra, Lemma 00L8. By Algebra, Lemma 00KC (which applies as R does not have finite length) the polynomial

$$n \longmapsto \text{length}_R(M/I^nM) - \text{length}_R(R/I^n) - \text{length}_R(Q/I^nQ)$$

has degree $< d$. Since $M = \prod R/\mathfrak{p}_i$ and since $n \to \text{length}_R(R/\mathfrak{p}_i + I^n)$ is a numerical polynomial of degree exactly(!) d for $i = 1, \ldots, t$ (by Algebra, Lemma 00L8) we see that the leading coefficient of $n \mapsto \text{length}_R(M/I^nM)$ is at least $t/d!$. Thus we conclude by Lemma 38261. □